How To Amend And Improve Tests And Code

Tutorial

Current Tests

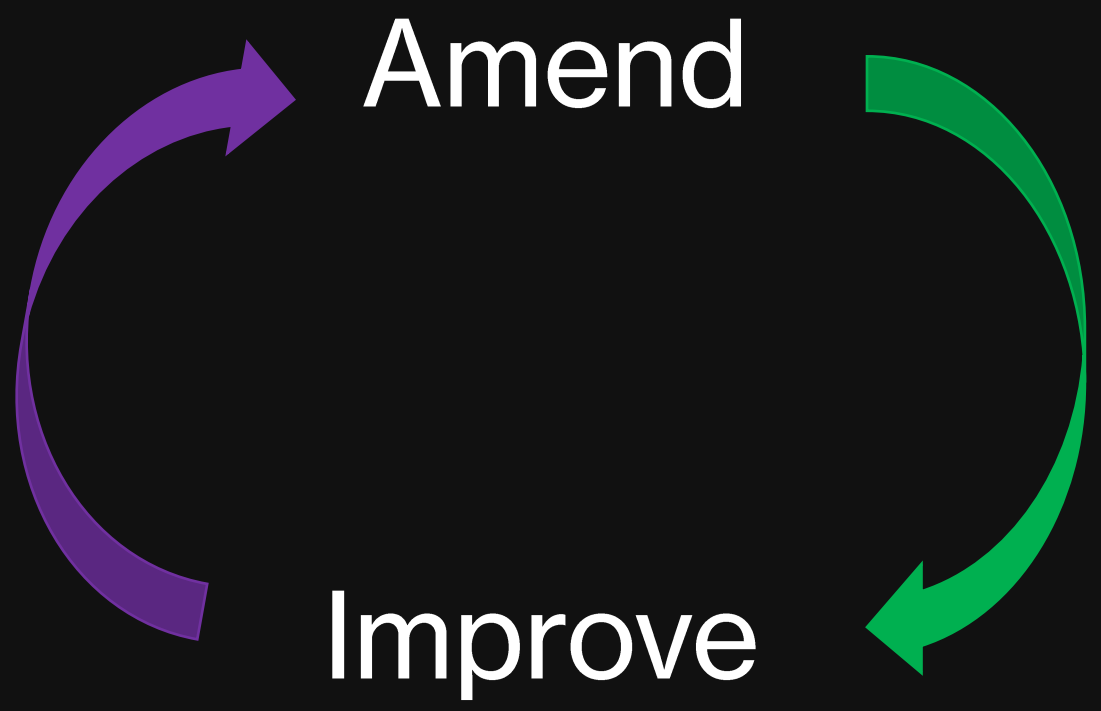
```
python -m pytest
========================================== test session starts ==========================================
[redacted]

collected 76 items

app\tests\backend\test_addtion.py ......                                                    [  7%]
app\tests\backend\test_calculator.py .....                                                  [ 14%]
app\tests\backend\test_division.py ......                                                   [ 22%]
app\tests\backend\test_error_handling.py ............................                       [ 60%]
app\tests\backend\test_input_handling.py ........                                           [ 71%]
app\tests\backend\test_multiplication.py ......                                             [ 78%]
app\tests\backend\test_output_handling.py ..........                                        [ 92%]
app\tests\backend\test_subtraction.py ......                                                [100%]

========================================== 76 passed in 0.29s ==========================================
```

Amend and Improve



Amend Tests

- Missing edge cases
  - Empty input values

```python
def test_handle_empty_input():
    with pytest.raises(ValueError) as error:
        handle_input("")
    assert str(error.value) == "Input cannot be empty"
```

Improve Tests

```python
def test_calculate_addition():
    res = calculate("6 + 6")
    assert res == "C"

def test_calculate_subtaction():
    res = calculate("6 - 6")
    assert res == "0"

def test_calculate_division():
    res = calculate("6 / 6")
    assert res == "1"

def test_calculate_multiplication():
    res = calculate("6 * 6")
    assert res == "24"

def test_calculate_expression():
    res = calculate_multiple("6 * 6 + 2")
    assert res == "26"
```

Improve Tests

```python
def test_calculate_addition():
    res = calculate("6 + 6")
    assert res == "C"
```

```python
def test_calculate_addition_success():
    res = calculate("6 + 6")
    assert res == "C"
```

Amend Code

- Missing edge cases
  - Empty input values

```
def handle_input(user_input):
    if (user_input == ""):
        raise ValueError("Input cannot be empty")
```

Improve Code

```python
def calculate(user_input):
    try:
        value_one, operator, value_two = handle_input(user_input)
        match operator:
            case "+":
                result = perform_addition(value_one, value_two)
            case "-":
                result = perform_subtraction(value_one, value_two)
            case "/":
                result = perform_division(value_one, value_two)
            case "*":
                result = perform_multiplication(value_one, value_two)

        final_result = handle_output(result)

        return final_result
    except Exception as e:
        raise
```

Next Step

Migrate Tests into GitHub Actions